CARNEGIE MELLON UNIVERSITY · SPRING 2027

11-769
Generative Machine Learning

Beyond Autoregressive LLMs

Six Paradigms. Three Data Domains.

Compare the strengths, limitations, and emerging applications of modern generative learning approaches — beyond a single modeling paradigm.

Autoregressive	**Diffusion**	**Flow Matching**
Energy-Based	**Consistency**	**Other Generative Models**

Language & Sequences	**Images & Continuous Signals**	**Graphs & Combinatorial Structures**

Who Should Take This Course?

NLP / LLM students

Explore diffusion, flow matching, and alternatives to autoregressive models.

ML students

Compare modern generative paradigms across data types and applications.

CSD students

Explore learning-based approaches to graphs and NP-hard problems.

OR / Economics students

Explore machine-learning approaches to optimization and structured decisions.

Prof. Yiming Yang

Language Technologies Institute · School of Computer Science

Graduate Course (12 units)
Spring 2027
Tue & Thu · 12:30–1:50 PM